



SAPIENZA
UNIVERSITÀ DI ROMA

Modellazione

MIP

How to work with the course

How the course is built, how every exercise is built and how the exam is built: the two-part path, the grading criteria, the typical discussion questions, the most common mistakes, and how every number quoted in the notes is reproduced.

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Online version and downloads:

fabiofurini.github.io/modellazione-mip/
github.com/fabiofurini/modellazione-mip

Academic year 2026–2027

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How to work with the course

The course runs in two parts: first you learn the modelling tools, then you apply them to complete problems. One thread holds them together — a model is not just written down: its optimum is enclosed between two numbers, $LB \leq z(MILP) \leq UB$, before it is ever handed to the solver.

1.1 The two-part path

	Content	Learning objectives
Part I	Modelling	recognise a link between variables (activation, maximum, big-M, if and only if...) and prove that the model really imposes it
Part II	The problems	apply the links to three families of real problems and to the mixed problems, from the model to Gurobi code

1.2 The format of every exercise (and of the exam)

Every problem of the course — and every exam question — follows the same four-question scheme:

1. **Model.** Write the MILP: variables (with their count), objective, constraints with their description, and — if two linked families of variables are involved — explain the link: what is the implication, and how the constraint (or the optimum) imposes it, in both directions.
2. **Instance.** Write the model for a small numerical instance.
3. **Heuristic.** Design a constructive algorithm that finds a feasible solution of the instance, and run it step by step to obtain an upper bound (lower bound if the problem is a maximisation).
4. **Dual.** Write the dual of the LP relaxation, for the general model and for the instance, and hand-build a feasible dual solution to obtain the bound from the other side.

The resulting $LB \leq z(MILP) \leq UB$ is the thread running through the course: a model is not just written down; its optimum is enclosed between two numbers before it is handed to the solver.

1.3 The additional questions and their solutions

Every problem closes with two or three *additional modelling questions*: they change a datum or add a constraint, and ask how the model changes and what the new optimum is. They are the part worth practising on: the model in the text is there to be read, the one for the variant has to be built.

The solutions of the additional questions are reserved for instructors: they live in a separate booklet, which is not published. What stays available to everyone — and it is what one needs in order to know what is expected — is the *sandwich worked out in full* on one variant of each exercise of chapters “Assignment and scheduling” and “Location and covering”: the model of the variant, a feasible constructive heuristic run step by step, a dual certificate built by hand and the bound table. It is the model answer the exam expects.

1.4 Grading criteria

Dimension	Weight
Correctness of the model (variables, objective, constraints)	35%
Proof of the link between the variables (both directions)	25%
Constructive heuristic and correct execution	20%
Dual of the relaxation and feasible dual solution	20%

1.5 Typical discussion questions

- Is the link constraint aggregated or disaggregated? What difference does it make to the LP relaxation?
- Is the opposite direction of the implication imposed by the constraint or does it follow from the optimum? How is it proved?
- What is the smallest big-M that can be justified from the data?
- Does the heuristic find the optimum? How can one know without the solver?
- Is the hand-built dual optimal for the relaxation, or only feasible?

1.6 The most common mistakes

1. Writing an implication whose thesis is true regardless (vacuous contrapositive): before writing it, check that antecedent and consequent are both genuine.
2. Proving only one direction of an implication “imposed by the constraint” — both “if”s are always needed.
3. Confusing a relation “imposed by the constraint” with one that “follows from the optimum”: the second requires the six-step exchange argument, not just “it’s clearly worth it”.
4. Concluding “in every optimal solution” when the coefficient is only ≥ 0 (not > 0): the correct conclusion is weaker, “there exists an optimum in which...”.
5. Using a huge big-M “to be safe”: it needlessly worsens the LP relaxation.

6. Forgetting that a mutual-exclusion constraint ($A + B \leq 1$) does not imply that at least one of the two must equal 1.
7. Writing the dual of the relaxation with the wrong direction or sign of a dual variable relative to the direction of the primal constraint.
8. Building a feasible dual solution without checking its feasibility on all the constraints.

1.7 Reproducibility

```
python3 -m pip install gurobipy matplotlib pandas
python3 python/run_all.py           # regenerates data, results, figures and
    notebooks
python3 python/check_numbers.py     # checks every number quoted in the notes
```

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